IMAGE TRANSFORMS

BY

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RESEARCH AND THEORY
HOW A VIEW MOVES, HOW WHAT IT WATCHES MOVES, AND HOW TO TELL THEM APART

ORIOR
2026

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For Megan, my loving wife, the only person that really understands what I mean when I speak.

For Autumn and Opal, without you this thought would have never occurred.

To everyone else, I wish you well.

ABSTRACT

The subject is the transform between two views of the same thing. An object tracked through a sequence of images, in any number of dimensions, is seen through a view that itself moves. What the images record is the object's motion and the view's motion added together, and a tracker that reads the sum as the object's motion follows the view instead of the object. This research paper collects the published ways a view's motion is recovered from the images alone, in the order of the transforms they recover: translation, rotation and scale, perspective, and flow over a curved surface that the tracked objects travel on.

What is quoted and what is new. Every method in the first four chapters is published and cited in the last, and is described here as its authors state it. Every one of them is computed in floating point: a fast Fourier transform, a least squares fit, an iterative solver stopped at a tolerance. The fifth chapter is the design this work adds, which carries the same transforms in exact arithmetic: integer fields, number theoretic transforms whose counts never reach their modulus, and rational rotations. Nothing in the fifth chapter has been measured here. It states what exactness is predicted to change. The predictions are on the page before any result is.

Why the order matters. A transform recovered first is removed before the next is looked for, and each removal leaves a residual that the next transform explains. Plane plus parallax states this for camera motion: register the dominant plane, and what remains is parallax plus the motion of whatever moved on its own. The same stratification holds for any subject that drifts, turns and deforms in front of a view that moves too, and it is the organizing idea of the research paper.

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1

OBSERVED MOTION IS TWO MOTIONS

1.1 THE SUM A SEQUENCE RECORDS

Let an object sit at position $p(t)$ in some frame of reference fixed to the thing being watched, and let the view map that frame to image coordinates by a transform T_t. The image records $x(t) = T_t(p(t))$, and the displacement between two frames is

$$x(t+1) - x(t) \;=\; \underbrace{T_{t+1}(p(t)) - T_t(p(t))}_{\text{the view moved}} \;+\; \underbrace{T_{t+1}(p(t+1)) - T_{t+1}(p(t))}_{\text{the object moved}}.$$

Nothing about an image separates the two terms. A tracker that links an object to whatever lies nearest its last image position is linking along the sum, and where the first term is as large as the second it is following the view.

Optical flow on a moving surface writes the same decomposition. Where the observed points lie on a surface that itself moves, the total velocity of a point moving along a level line of constant intensity is

$$\hat{\mathbf{M}} \;=\; \hat{\mathbf{V}} + \hat{\mathbf{v}},$$

where $\hat{\mathbf{V}}$ is the velocity of the surface and $\hat{\mathbf{v}}$ is the tangential velocity of the points relative to it. The flow is posed for $\hat{\mathbf{v}}$ alone, once the surface velocity is fixed by a choice of parametrization.

1.2 PLANE PLUS PARALLAX

For a camera moving through a three dimensional scene the stratification is published and precise [2]. Register the images by the homography of one dominant plane. That single projective transform removes every effect of camera rotation, zoom and calibration without computing any of them explicitly. What remains has two sources only: parallax, which the camera's translation produces at points off the plane, and objects that moved on their own. The residual parallax field is epipolar, radial about the epipole. A residual that is not radial about that point is independent motion.

Irani and Anandan [8] used this to stratify moving object detection from two dimensional scenes up to full three dimensional ones, and extended the recovery of planar parallax to many frames at once. Triggs [18] related the plane plus parallax representation to the tensors of multiple view geometry.

1.3 THE ORDER TRANSFORMS ARE REMOVED IN

The lesson carried through this research paper is an order. Remove the transform of the whole view first, because every object shares it. Then remove what a group of objects shares, a medium flowing or a surface turning under them. The remainder is each object's own motion, and a tracker should link on that motion alone. Each stage explains part of the residual of the stage before it, and each stage that is skipped leaves its motion inside every object's apparent step.

Two consequences follow before any method is chosen. First, the view's motion is estimated from everything in the image, not from the objects being tracked, because everything in the image shares it. Second, a view that orbits its subject while the subject moves changes perspective as well as position. A translation alone cannot be the whole first stage.

2

TRANSLATION

2.1 AGREEMENT AT EVERY LAG

For two views a and b of the same extents, the agreement at a lag ℓ is

$$C(\ell) \;=\; \sum_{v} a(v)\, b(v+\ell),$$

and the lag with the largest agreement is the translation between them. Computed lag by lag this costs the volume times the number of lags. Computed through the convolution theorem it costs one transform of each view, one pointwise product, and one inverse transform, for every lag at once.

2.2 PHASE CORRELATION

Phase correlation rests on the Fourier shift theorem [21]. If two images differ by a displacement $(\Delta x, \Delta y)$, their Fourier transforms are related by

$$I_2(\omega_x, \omega_y) \;=\; I_1(\omega_x, \omega_y)\, e^{-j(\omega_x \Delta x + \omega_y \Delta y)},$$

so the normalized cross power spectrum is a pure phase whose inverse transform is a delta at the displacement. The method extends to N dimensions directly [4]. Under translation the motion model for N dimensional data is a rank one tensor, and a higher order singular value decomposition of the phase correlation separates the displacement along each dimension independently.

Sub-pixel precision is recovered from the shape of the peak. The peak of phase correlation is a Dirichlet kernel, closely approximated by a sinc, and fitting a sinc, a Gaussian or a triangle to it recovers the fraction of a pixel. Foroosh, Zerubia and Berthod [7] derive the sub-pixel extension in closed form. An upsampled matrix multiplication of the discrete Fourier transform gives arbitrary sub-pixel precision without transforming a larger grid, and is how the widely used `register_translation` of scikit-image [17] works.

2.3 DRIFT CORRECTION

Time-lapse imaging uses exactly this to remove the drift of a subject in its view. In published microscopy protocols phase correlation registers each time point of a four dimensional recording [12] while keeping the movements of the objects inside it measurable, and in long recordings

the global movement of the subject is estimated during acquisition and fed back to the stage so the subject stays in the field of view. Video stabilization does the same for a handheld camera. In every case the translation of the whole view is removed before any object is tracked, the order the first chapter states.

2.4 WHERE TRANSLATION STOPS

A translation has as many parameters as the view has axes and explains only the motion every point shares equally. A view that turns, zooms or changes perspective moves points by different amounts in different places, and its residual after the best translation is a structured field instead of scatter. The next chapter takes those transforms in turn.

3

ROTATION, SCALE AND PERSPECTIVE

3.1 ROTATION AND SCALE AS TRANSLATIONS

The magnitude of a Fourier transform does not change when the image translates, and for this reason it is called the Fourier-Mellin invariant. Resample that magnitude on a log-polar grid and the two remaining similarity parameters become translations: a rotation of the image is a translation along the angle axis, and a uniform scaling by a constant factor is a translation along the log radius axis. Phase correlation in log-polar space recovers rotation and scale, the image is corrected for both, and phase correlation in the original space then recovers the translation. Reddy and Chatterji [15] give the FFT based technique; later work adds pseudo-polar transforms and robustness to noise [5]. Log-polar resampling in the spatial domain also recovers large scale changes, at least four-fold, and arbitrary rotations as a preprocessing step.

3.2 ROTATION IN THREE DIMENSIONS

A turning specimen in a volume is a rotation in the group $SO(3)$, and the log-polar trick does not carry over because $SO(3)$ is not a product of circles. Two published routes do.

Through projections. The projection-slice theorem states that the Fourier transform of a projection of a volume is a central slice of the volume's Fourier transform. Combined with phase correlation it estimates rotation parameters from projections, including perspective projections and in the presence of translational offsets, and it widens the capture range of iterative registration substantially: in one clinical evaluation initialized registration succeeded in 68.6 percent of cases against 28.8 percent uninitialized [19].

Through spherical harmonics. Makadia and Daniilidis [10] estimate rotation directly from functions on the sphere, without correspondences. Spherical harmonic coefficients undergo a unitary map when the function is rotated. The correlation of two spherical functions is a function on $SO(3)$, and its $SO(3)$ Fourier transform is the pointwise product of the two functions' spherical harmonic coefficients. The rotation is the peak of that correlation, found by a direct search over a grid on $SO(3)$ whose resolution is set by the harmonic bandwidth. The same construction registers three dimensional point clouds [11].

A space of spherical harmonics of degree n has dimension $2n + 1$, and its members are eigenfunctions of the Laplace-Beltrami operator with eigenvalue $n(n + 1)$. Degree 1 is where a rigid turn of the whole sphere lives. Rotation is therefore read off the lowest degrees and finer motion off higher ones.

3.3 PERSPECTIVE

A homography is a projective transform. It maps points between two views exactly in two cases: a planar scene seen under arbitrary camera motion, and an arbitrary scene seen under pure camera rotation. It preserves collinearity and not parallelism, length or angle. With four point correspondences the direct linear transform [20] solves for it as a linear system, and random sample consensus makes the estimate robust to wrong correspondences.

Without correspondences the warp is estimated directly from intensities. Lucas and Kanade's alignment, reviewed in a unifying framework by Baker and Matthews [3], treats translation, affine and homography warps alike: it linearizes the image around the current warp and updates the parameters by gradient descent. The inverse compositional form precomputes the expensive terms once, on the template, and is the efficient variant most extensions can use without losing that efficiency.

3.4 RIGID REGISTRATION FROM POINT CONSTELLATIONS

Where views differ by a large rotation, intensity based alignment loses its capture range and point based registration keeps it. One published method [13], built for multi-view volumetric microscopy, embeds fiducial points and matches their constellations with a local geometric descriptor invariant to translation and rotation, reaching an average displacement of about one pixel. Because it depends only on repeatable point detections it is more robust than intensity based registration and orders of magnitude faster, and the same software is used to stabilize three dimensional time series as well as to align views.

4

WAVES ON A SURFACE

4.1 OBJECTS THAT RIDE A MOVING SURFACE

Many tracked objects do not move freely through a volume. They travel on a surface: tracers on a fluid interface, weather systems on a planet, features on a deforming membrane, layers of cells on a developing organism. When that surface is close to a sphere it is modeled as an evolving sphere-like surface, a closed surface $\mathscr{M}_t$ reached from the unit sphere by a radial map,

$$x \;\mapsto\; \tilde{\rho}(t,x)\,x, \qquad x \in \mathcal{S}^2,$$

so motion estimation on the surface is posed on the round sphere once the radius $\tilde{\rho}$ is known. Lang and Scherzer [9] set this up for optical flow and give the full construction; this chapter follows theirs.

4.2 FITTING THE SURFACE

The radius is fitted to sample points on the surface, the approximate centers of the tracked objects, after centring them on a fitted sphere. The fit minimizes a data term plus a Sobolev seminorm penalty,

$$\mathscr{F}_\beta(\tilde{\rho}) \;=\; \|\tilde{\rho} - \tilde{\rho}^{\delta}\|^2_{L^2(\mathcal{S}^2)} + \beta\,|\tilde{\rho}|^2_{H^s(\mathcal{S}^2)},$$

with the radius expanded in scalar spherical harmonics. The seminorm is diagonal in the harmonic basis, weighting degree n by $(n(n+1))^s$. Smoothness is a statement about how fast coefficients decay with degree. Image values on the surface are read as the maximum intensity along a narrow radial band about it.

4.3 THE FLOW IN VECTOR SPHERICAL HARMONICS

Every tangent vector field on the sphere is expanded uniquely in tangential vector spherical harmonics of two types,

$$\tilde{\mathbf{y}}^{(2)}_{nj} = \lambda_n^{-1/2}\,\nabla_{\mathcal{S}^2}\tilde{Y}_{nj}, \qquad \tilde{\mathbf{y}}^{(3)}_{nj} = \lambda_n^{-1/2}\,\nabla_{\mathcal{S}^2}\tilde{Y}_{nj} \times \tilde{\mathbf{N}},$$

with $\lambda_n = n(n+1)$. The first type is a gradient field and carries the part of a flow that converges or diverges; the second is a rotated gradient and carries the part that circulates. The expansion is therefore a Helmholtz-Hodge decomposition already, and vector spherical harmonics are called

the natural basis for it [1]: a flow on a sphere splits uniquely into an irrotational part, a solenoidal part, and a harmonic part. There is no tangential harmonic of degree zero, since by the hairy ball theorem no non-vanishing tangent field is constant over the sphere.

The optical flow on the surface minimizes a data term, the squared residual of the generalized optical flow equation, plus α times the squared H^1 norm of the field. Expanded in vector harmonics up to a chosen degree, it becomes a linear system $(A + \alpha D)v = b$ solved iteratively to a tolerance. The integrals are evaluated on a refined icosahedron with piecewise quadratic interpolation and a quadrature rule per triangle, and in the published implementation those integrals, not the linear solve, dominate the run time.

4.4 WHAT THE HARMONICS BUY A TRACKER

The decomposition is ordered by scale. Degree 1 of the circulating type is a rigid turn of the whole surface, the motion a turning view or a turning subject produces everywhere at once. Low degrees carry the large-scale flows of the surface, the waves that move whole neighborhoods of objects together. What no low degree explains is the motion particular to one object. A tracker that removes the low degrees before it links is linking on that remainder, the order the first chapter asks for.

5

THE SAME TRANSFORMS IN EXACT ARITHMETIC

One transform here is demonstrated; the other two are design only, and the predictions are unmeasured. The translation by number theoretic transform is built and run: its correlation agrees to the digit with the direct $O(N^2)$ count and recovers the planted shift. That is a run against a reference and not a prediction, and the demonstration is in the orior repository, `https://github.com/dstroy0/orior`. The precision constants it rests on are re-derived by certificate, and a wrong one is refused [14]. The integer-field and rotation transforms are not built: nothing in their sections has been measured, and they are the design this work adds to the methods quoted before it, stated before any result. The predictions at the end of the chapter concern behavior on real sequences and remain unmeasured for every transform, the translation included.

5.1 WHY EXACTNESS CHANGES THE ORDER OF OPERATIONS

Every published method in this research paper rounds. A fast Fourier transform carries each coefficient to machine precision, a sub-pixel peak is fitted by a curve, a variational problem is solved to a tolerance, and a spherical integral is a quadrature on a mesh. Each rounding is small, and each is fed to the next stage: the translation that was rounded is removed before the rotation is estimated, and the rotation that was rounded is removed before the flow is. A stratified pipeline is the arrangement in which small errors compound, because every stage reads the residual the last stage left, including what the last stage got wrong.

The consequence for design is specific. Averaging exists to cancel error: fitting a line across many frames, pooling neighbors, smoothing a field. Where the arithmetic makes no error there is nothing to cancel, and averaging only mixes the present with the past and one object with another. What remains useful is accumulation: exact counts added over time, which lose nothing however many are added.

5.2 INTEGER FIELDS

A smoothing kernel with integer weights keeps a field in the integers. The binomial kernel of even order n, the nth row of Pascal's triangle, sums to exactly 2^n and has variance $n/4$ in units of the grid spacing. It is the integer counterpart of a Gaussian of that variance. Separable along each axis, a volume smoothed by orders $(n_1, \dots, n_d)$ is scaled by exactly $2^{n_1+\cdots+n_d}$, and the difference of two smoothings,

$$r \;=\; s \cdot 2^{\sum_i m_i} \;-\; b,$$

with s smoothed by orders n_i and b smoothed further by orders m_i, compares them at one common scale with nothing divided. Its width is fixed by the orders and the input depth. It is carried in a fixed number of limbs, and its sign says exactly whether a point stands above its local background.

5.3 TRANSLATION BY NUMBER THEORETIC TRANSFORMS

The agreement at every lag, $C(\ell) = \sum_v a(v)\, b(v + \ell)$, is a convolution of the reflected first view with the second. A number theoretic transform [6, 16] computes a convolution exactly in the integers modulo a prime p that carries roots of unity of the transform length. The prime is $p = 119 \cdot 2^{23} + 1 = 998244353$, with $p - 1 = 2^{23} \cdot 7 \cdot 17$. 3 is a primitive root and the longest power-of-two transform the modulus admits has length 2^{23}, past any image axis. The prime, its generator and the 2^{27} device primes beside it are proved, each with a certificate a reader re-checks in one modular exponentiation [14]. This prime is below 2^{31}, which the same proof shows is the property that keeps a butterfly's sum and its subtraction from wrapping in a 32-bit word; a modulus above 2^{31} carries a silent wrong answer there. A single small prime keeps this transform clear of that boundary.

Exactness has a stated bound. Each coefficient is a sum of products $a(v)\, b(v+\ell)$. For binary views it counts agreeing positions and is at most the transform length. The transform length divides $p - 1$ and so is below p for any valid prime, and a binary view is then exact with no separate precondition: no residue is ever reduced. Weighting the views can breach p. The bound becomes the sum of the weight products, and past p a single prime counts modulo itself, silently, the same fault one level up from a wrong constant. The remedy is a prime above the largest coefficient, or the Chinese remainder theorem over several, the device path of that proof reassembling a 94-bit product. A view is embedded on each axis in a period of at least twice its extent less one so that no two lags alias, and the transform is separable, one axis at a time. The construction holds in any number of dimensions.

The peak is then an integer, and ties among lags are broken by a rule instead of by rounding: the shortest lag by a weighted squared length whose weights carry the view's anisotropy, then a fixed index order.

5.4 ROTATION WITHOUT ANGLES

A rotation in the plane is multiplication by a unit complex number, and the ratio of two directions is a complex number whose argument is the turn between them. Where directions are given by rational coordinates the ratio is a Gaussian rational, and applying it is exact. A turn carried forward one step is exact in the same way. What is not rational is a fraction of a turn, a cube root of a rotation for instance, and a design that needs one has left exact arithmetic and should say so.

For turns in three dimensions the same holds for quaternions with rational components. The spherical harmonic route of the third chapter needs coefficients that are integrals, and a function sampled at points gives those integrals only through a quadrature rule; which quadratures are exact for which degrees on which point sets is the question an exact design has to answer first.

5.5 PREDICTIONS

First. Removing the view's motion before linking will reduce links made to the wrong object more than any change to the linker does, wherever the view moves by as much as the objects do, because in that regime most of every object's apparent step belongs to the view.

Second. A view that orbits its subject leaves a residual after the best translation that is structured instead of scattered: parallax that varies with depth, or a turn that varies with distance from an axis. The residual of the translation stage is the measurement that says the next transform is needed.

Third. Each transform recovered exactly and removed will leave the next stage a residual with no rounding in it. The improvement from adding a stage will not shrink the way it does in a floating point pipeline, where each stage inherits the error of the last.

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